DATA SCIENCE

Data Analyst to Data Scientist in 6 Months. My Exact Roadmap

A month-by-month roadmap for working data analysts moving into data science. Builds on SQL and Excel you already know.

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What you already have

An analyst is not starting from zero. Use what you already know.

- **Already strong**

  SQL, data cleaning, business questions, dashboards, explaining numbers.

- **The gap to close**

  Python for modelling, statistics, machine learning, and evaluating models.

- **Your edge**

  Data scientists who cannot explain results are common. You already can.

Time: 1.5 hours on weekdays, 3-4 hours on weekends, alongside your current job.

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Month 1: Python for data science

Get as comfortable in Python as you are in SQL.

- **Weeks 1-2**

  Pandas in depth: groupby, merge, pivot, dates, missing values. Redo 10 of your SQL analyses in Pandas.

- **Week 3**

  NumPy basics and charts with Matplotlib or Seaborn.

- **Week 4**

  Write clean functions and notebooks. Mini project: a full exploratory analysis with findings.

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Month 2: Statistics

The part most analysts skip, and interviewers test the most.

- **Week 1**

  Distributions, sampling, Central Limit Theorem.

- **Week 2**

  Hypothesis testing, p-values, confidence intervals.

- **Week 3**

  A/B testing end to end: sample size, running, reading results.

- **Week 4**

  Correlation, linear regression and how to read coefficients.

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Months 3 and 4: Machine learning

Learn the core algorithms with scikit-learn on real data.

Month 3

Train-test split, linear and logistic regression, decision trees, random forest, metrics (MAE, precision, recall, F1).

Month 4

Gradient boosting, cross-validation, feature engineering, imbalanced data, pipelines, K-Means.

Rule

For every algorithm, build it on one dataset and explain the result in business terms.

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Month 5: Two projects

Pick problems from your current domain. That is your strongest story.

- **Project 1: Classification**

  Example: predict which customers will churn. Show recall, top drivers and the ₹ value of acting on it.

- **Project 2: Regression or forecasting**

  Example: forecast next quarter's demand by region and compare against a simple baseline.

- **For both**

  GitHub repo, clean notebook, a README with the business problem, approach, result and limitations.

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Month 6: Make the switch

The fastest route is often inside your current company.

- **Internal move**

  Offer a small model that helps your current team. A working model is the best application.

- **Resume**

  Title your projects by business outcome: "Churn model that flags 70% of churners 30 days early".

- **Interview prep**

  Statistics questions, ML concepts, SQL, and walking through both projects.

- **Target roles**

  Data scientist, product analyst with ML, decision scientist, analytics engineer with ML.

Honest note: 6 months makes you ready to apply for junior and associate data scientist roles. Interviews and the move itself can take a few more months.

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